Earthquake Analysis Dashboard using Python

Interactive Visualization of Global Seismic Activity

Domain: Remote Sensing & GIS

Project Report

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Abstract

Earthquakes are among the most significant natural hazards, making the visualization and analysis of seismic activity essential for researchers and decision-makers. This project presents an interactive Earthquake Analysis Dashboard developed using Python for interactive geospatial visualization and statistical analysis.The dashboard automatically retrieves the latest global earthquake data from the United States Geological Survey (USGS), performs basic statistical analysis, and visualizes earthquake locations on an interactive world map. The project also generates summary statistics such as the total number of earthquakes, average magnitude, strongest earthquake, and deepest earthquake, along with graphical representations of magnitude distribution. The proposed dashboard demonstrates how publicly available geospatial data can be transformed into meaningful visualizations using open-source tools such as Pandas, Folium, and Matplotlib. This project serves as a simple and effective example of integrating GIS and Python for real-time geospatial data analysis.

Introduction

Earthquakes occur due to the sudden release of energy within the Earth's crust, generating seismic waves that can cause significant damage to life and infrastructure. Monitoring global seismic activity helps scientists understand tectonic processes and identify regions with frequent earthquake occurrences.

With the increasing availability of open geospatial datasets, it has become possible to develop interactive applications for visualizing and analyzing earthquake information. This project utilizes publicly available earthquake data from the United States Geological Survey (USGS) and processes it using Python-based data analysis and GIS libraries.

The primary objective of this project is to demonstrate how real-time seismic data can be collected, analyzed, and presented through an interactive dashboard. The project emphasizes ease of use, reproducibility, and the application of basic GIS concepts for spatial data visualization.

Objectives

The primary objectives of this project are:

- To collect real-time global earthquake data from the USGS Earthquake Catalog.
- To perform basic statistical analysis of recent earthquake events.
- To visualize earthquake locations on an interactive world map using GIS techniques.
- To analyze the distribution of earthquake magnitudes using graphical representation.
- To demonstrate the application of Python and open-source GIS libraries for geospatial data analysis.

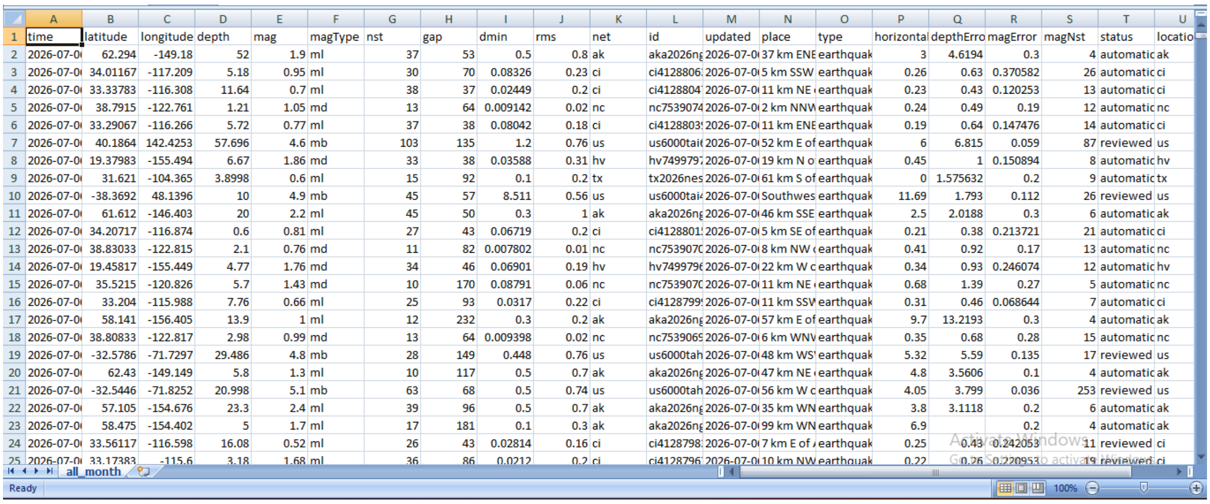
Methodology

The project follows a simple workflow for earthquake data analysis:

1. **Data Collection:** Recent global earthquake data is fetched from the USGS Earthquake Catalog in CSV format.
2. **Data Processing:** The dataset is loaded into a Pandas DataFrame for cleaning and analysis.
3. **Statistical Analysis:** Basic statistics such as total earthquakes, average magnitude, strongest earthquake, and deepest earthquake are computed.
4. **Interactive Mapping:** The Folium library is used to plot earthquake locations on an interactive world map. Marker colors represent earthquake magnitude, and clicking a marker displays additional information.
5. **Visualization:** A histogram is generated to study the distribution of earthquake magnitudes, and the top ten strongest earthquakes are identified from the dataset.

Tools and Technologies

Tool / Library	Purpose
Google Colab	Development environment
Python	Programming language
Pandas	Data collection and analysis
Folium	Interactive GIS map visualization
Matplotlib	Statistical visualization
USGS Earthquake Catalog	Source of real-time earthquake data



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	time	latitude	longitude	depth	mag	magType	nst	gap	dmin	rms	net	id	updated	place	type	horizontal	depthErro	magError	magNst	status	locatio
2	2026-07-0	62.294	-149.18	52	1.9	ml	37	53	0.5	0.8	ak	aka2026n	2026-07-0	37 km ENE	earthquak	3	4.6194	0.3	4	automatic	ak
3	2026-07-0	34.01167	-117.209	5.18	0.95	ml	30	70	0.08326	0.23	ci	ci4128806	2026-07-0	5 km SSW	earthquak	0.26	0.63	0.370582	26	automatic	ci
4	2026-07-0	33.33783	-116.308	11.64	0.7	ml	38	37	0.02449	0.2	ci	ci4128804	2026-07-0	11 km NE	earthquak	0.23	0.43	0.120253	13	automatic	ci
5	2026-07-0	38.7915	-122.761	1.21	1.05	md	13	64	0.009142	0.02	nc	nc7539074	2026-07-0	2 km NNW	earthquak	0.24	0.49	0.19	12	automatic	nc
6	2026-07-0	33.29067	-116.266	5.72	0.77	ml	37	38	0.08042	0.18	ci	ci4128803	2026-07-0	11 km ENE	earthquak	0.19	0.64	0.147476	14	automatic	ci
7	2026-07-0	40.1864	142.4253	57.696	4.6	mb	103	135	1.2	0.76	us	us6000tai	2026-07-0	52 km E of	earthquak	6	6.815	0.059	87	reviewed	us
8	2026-07-0	19.37983	-155.494	6.67	1.86	md	33	38	0.03588	0.31	hv	hv7499797	2026-07-0	19 km N o	earthquak	0.45	1	0.150894	8	automatic	hv
9	2026-07-0	31.621	-104.365	3.8998	0.6	ml	15	92	0.1	0.2	tx	tx2026nes	2026-07-0	61 km S of	earthquak	0	1.575632	0.2	9	automatic	tx
10	2026-07-0	-38.3692	48.1396	10	4.9	mb	45	57	8.511	0.56	us	us6000tai	2026-07-0	Southwes	earthquak	11.69	1.793	0.112	26	reviewed	us
11	2026-07-0	61.612	-146.403	20	2.2	ml	45	50	0.3	1	ak	aka2026n	2026-07-0	46 km SSE	earthquak	2.5	2.0188	0.3	6	automatic	ak
12	2026-07-0	34.20717	-116.874	0.6	0.81	ml	27	43	0.06719	0.2	ci	ci4128801	2026-07-0	5 km SE of	earthquak	0.21	0.38	0.213721	21	automatic	ci
13	2026-07-0	38.83033	-122.815	2.1	0.76	md	11	82	0.007802	0.01	nc	nc753907	2026-07-0	8 km NW	earthquak	0.41	0.92	0.17	13	automatic	nc
14	2026-07-0	19.45817	-155.449	4.77	1.76	md	34	46	0.06901	0.19	hv	hv7499796	2026-07-0	22 km W c	earthquak	0.34	0.93	0.246074	12	automatic	hv
15	2026-07-0	35.5215	-120.826	5.7	1.43	md	10	170	0.08791	0.06	nc	nc753907	2026-07-0	11 km NE	earthquak	0.68	1.39	0.27	5	automatic	nc
16	2026-07-0	33.204	-115.988	7.76	0.66	ml	25	93	0.0317	0.22	ci	ci4128799	2026-07-0	11 km SSW	earthquak	0.31	0.46	0.068644	7	automatic	ci
17	2026-07-0	58.141	-156.405	13.9	1	ml	12	232	0.3	0.2	ak	aka2026n	2026-07-0	57 km E of	earthquak	9.7	13.2193	0.3	4	automatic	ak
18	2026-07-0	38.80833	-122.817	2.98	0.99	md	13	64	0.009398	0.02	nc	nc7539069	2026-07-0	6 km WNW	earthquak	0.35	0.68	0.28	15	automatic	nc
19	2026-07-0	-32.5786	-71.7297	29.486	4.8	mb	28	149	0.448	0.76	us	us6000tah	2026-07-0	48 km WS	earthquak	5.32	5.59	0.135	17	reviewed	us
20	2026-07-0	62.43	-149.149	5.8	1.3	ml	10	117	0.5	0.7	ak	aka2026n	2026-07-0	47 km NE	earthquak	4.8	3.5606	0.1	4	automatic	ak
21	2026-07-0	-32.5446	-71.8252	20.998	5.1	mb	63	68	0.5	0.74	us	us6000tah	2026-07-0	56 km W c	earthquak	4.05	3.799	0.036	253	reviewed	us
22	2026-07-0	57.105	-154.676	23.3	2.4	ml	39	96	0.5	0.7	ak	aka2026n	2026-07-0	35 km WN	earthquak	3.8	3.1118	0.2	6	automatic	ak
23	2026-07-0	58.475	-154.402	5	1.7	ml	17	181	0.1	0.3	ak	aka2026n	2026-07-0	99 km WN	earthquak	6.9		0.2	4	automatic	ak
24	2026-07-0	33.56117	-116.598	16.08	0.52	ml	26	43	0.02814	0.16	ci	ci4128798	2026-07-0	7 km E of	earthquak	0.25	0.43	0.242053	11	reviewed	ci
25	2026-07-0	33.17383	-115.6	3.18	1.68	ml	36	86	0.0212	0.2	ci	ci4128796	2026-07-0	10 km NW	earthquak	0.22	0.26	0.220953	19	reviewed	ci

Fig. 1: USGS data preview

Results and Discussion

The dashboard successfully retrieved live earthquake data from the United States Geological Survey (USGS) and performed basic statistical analysis.

The dataset contained **10,529 earthquake records** representing recent global seismic activity. The calculated **average earthquake magnitude** was **1.67**, indicating that most recorded earthquakes during the selected period were of relatively low intensity.

The strongest earthquake recorded in the dataset had a **magnitude of 7.8**, occurring **25 km southwest of Kablalan, Philippines**. The deepest earthquake reached a depth of **685.692 km** and occurred **274 km southeast of Levuka, Fiji**.

The interactive GIS map displayed the spatial distribution of earthquakes worldwide, allowing users to explore individual events through clickable markers containing location, magnitude, depth, and occurrence time.

The magnitude distribution histogram showed that the majority of earthquakes had magnitudes below 3, while high-magnitude earthquakes were comparatively rare. This observation is consistent with the general global distribution of seismic events, where smaller earthquakes occur much more frequently than larger ones.

Overall, the dashboard demonstrates that combining real-time geospatial data with Python-based analysis provides an effective way to understand global earthquake activity through both statistical summaries and interactive visualization.

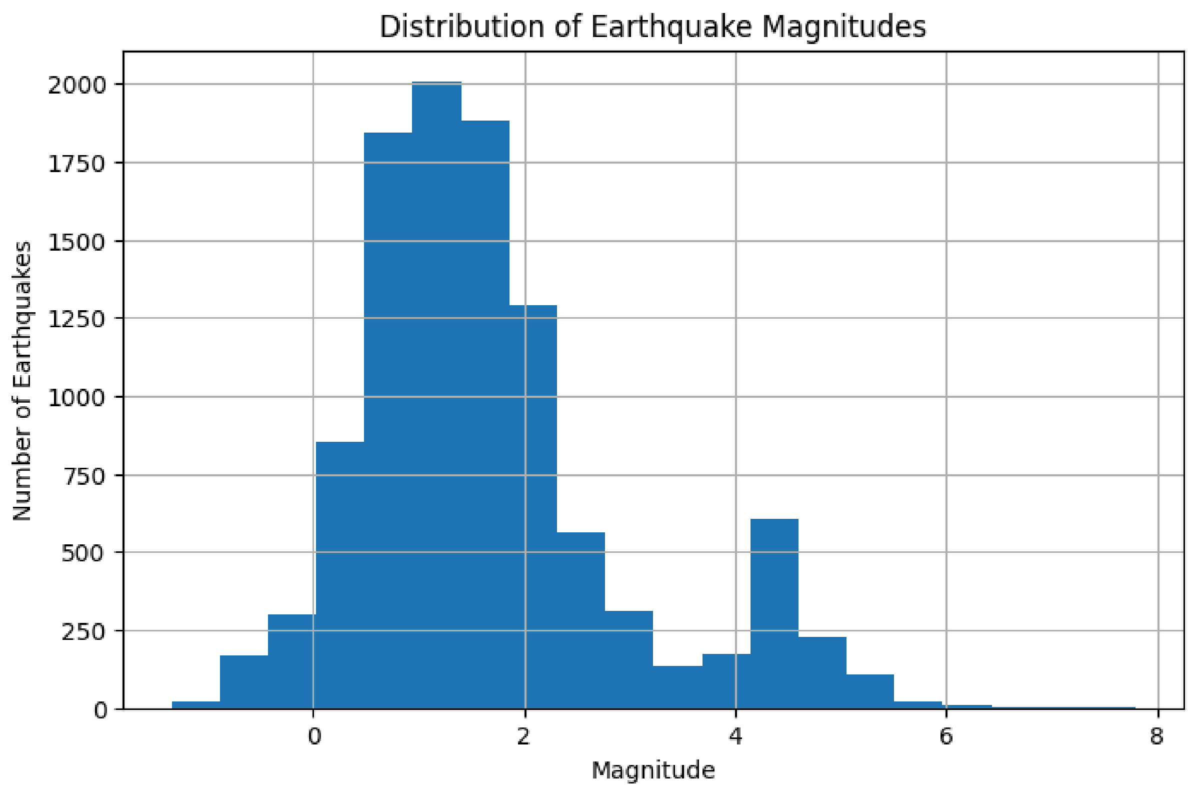


Fig. 2: Distribution of earthquake magnitudes.

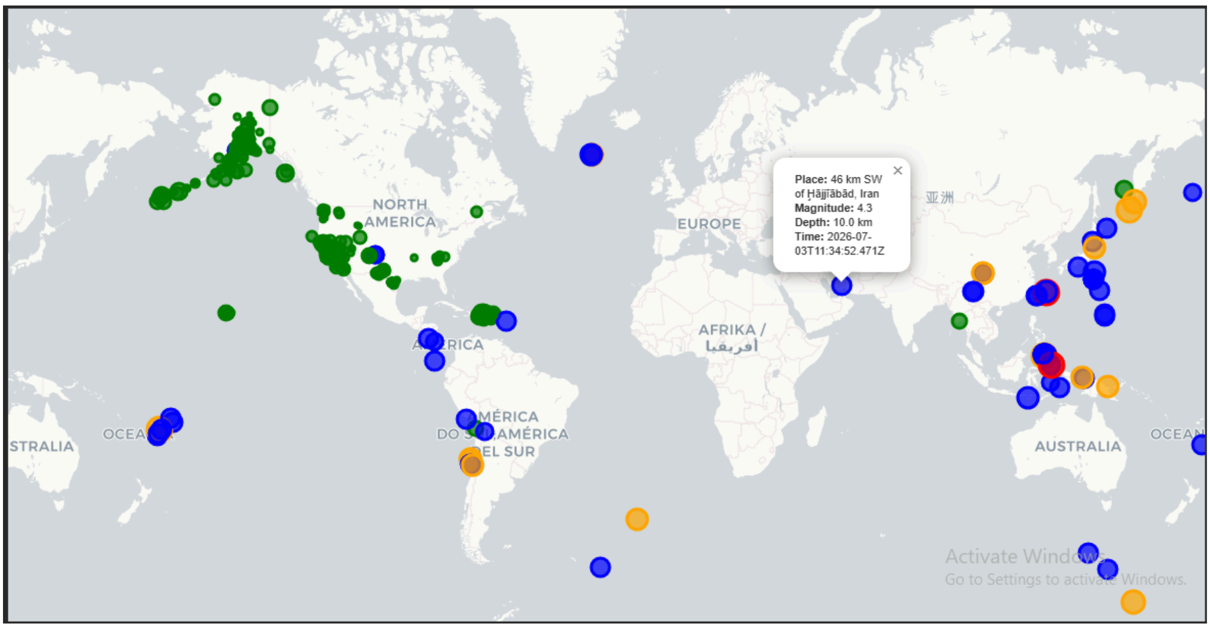


Fig. 3: Interactive visualization of recent global earthquakes.

index	place	mag	depth	time
10000	25 km SW of Kablalan, Philippines	7.8	56.991	2026-06-07T23:37:41.970Z
3273	20 km ESE of Yumare, Venezuela	7.5	10.0	2026-06-24T22:05:11.743Z
3274	21 km ENE of San Felipe, Venezuela	7.2	27.588	2026-06-24T22:04:34.363Z
3269	33 km ENE of Noda, Japan	6.9	34.0	2026-06-24T22:30:12.989Z
6697	43 km ESE of Palu, Indonesia	6.7	10.0	2026-06-16T03:27:44.418Z
5963	central Mid-Atlantic Ridge	6.6	10.0	2026-06-17T18:56:58.028Z
5426	133 km ESE of Petropavlovsk-Kamchatsky, Russia	6.6	10.0	2026-06-19T06:52:31.597Z
9974	18 km SW of Balangonan, Philippines	6.5	75.065	2026-06-08T00:55:12.546Z
2781	34 km WSW of Sarangani, Philippines	6.5	42.0	2026-06-26T11:34:41.498Z
6596	262 km SSE of Dunhuang, China	6.3	10.0	2026-06-16T09:06:55.517Z

Fig. 4: Top 10 strongest earthquakes in the dataset.

Conclusion

This project demonstrates a simple yet effective approach for analyzing and visualizing global earthquake data using Python. By integrating real-time earthquake data from the USGS with open-source libraries such as Pandas, Folium, and Matplotlib, the dashboard provides both statistical insights and interactive spatial visualization.

The project highlights how publicly available geospatial datasets can be transformed into meaningful information through basic data analysis and visualization techniques. Although designed as a student project, the developed dashboard illustrates the practical application of GIS concepts in understanding global seismic activity and serves as a foundation for more advanced earthquake monitoring systems.

Future Scope

The project can be extended in several ways:

- Integrate real-time automatic updates at regular intervals.
- Add earthquake prediction or risk assessment models using machine learning.
- Develop a web-based dashboard with advanced filtering and search features.
- Display tectonic plate boundaries and fault lines for improved geological context.
- Incorporate historical earthquake data to analyze long-term seismic trends.
- Generate automatic alerts for earthquakes above a user-defined magnitude threshold.

References

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5. Google Colab Documentation. https://colab.research.google.com/